

BOSTOVO CHEREPOVETS, Russian Federation

WMO: 271130

Lat: **59.2826N**Lon: **38.0256E**Elev: **117**StdP: **99.92**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.4	-26.1	-32.2	0.2	-29.2	-28.7	0.3	-26.0	9.0	-4.2	8.2	-6.1	0.8	50	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	28.2	20.0	26.3	19.1	24.4	18.1	21.2	26.4	20.1	24.8	19.0	23.1	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.3	14.3	23.9	18.3	13.4	22.4	17.3	12.6	21.3	61.9	26.6	58.1	24.9	54.4	23.1	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.2	6.1	-33.5	30.7	3.7	1.9	-36.1	32.0	-38.2	33.1	-40.3	34.2	-42.9	35.5		
(5)			-33.6	22.8	3.6	1.5	-36.2	23.8	-38.3	24.7	-40.3	25.6	-43.0	26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-10.0	-9.3	-4.1	3.6	10.6	14.7	17.7	15.1	10.0	3.8	-2.5	-6.8
	DBStd	11.02	8.01	7.91	5.39	4.48	4.71	3.92	3.55	3.42	3.75	4.17	5.85	7.58
	HDD10.0	2991	619	541	438	198	51	5	1	2	45	195	374	522
	HDD18.3	5426	877	774	696	443	243	121	56	111	249	451	624	781
	CDD10.0	667	0	0	0	6	69	147	238	160	46	2	0	0
	CDD18.3	62	0	0	0	0	3	13	35	11	0	0	0	0
	CDH23.3	648	0	0	0	1	48	156	326	116	2	0	0	0
	CDH26.7	150	0	0	0	0	7	34	84	25	0	0	0	0
Wind	WSAvg	2.6	2.7	2.8	2.8	2.6	2.8	2.5	2.2	2.2	2.3	2.8	2.9	3.0
Precipitation	PrecAvg	890	88	67	60	44	58	77	82	79	69	81	93	93
	PrecMax	1150	137	120	111	82	104	132	149	165	125	153	168	144
	PrecMin	687	36	18	11	19	17	18	7	20	37	28	12	41
	PrecStd	108	25	26	22	17	19	26	40	35	23	32	36	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	3.5	9.6	21.6	27.8	29.9	31.8	29.5	23.3	16.7	8.7	6.2
		MCWB	1.8	1.9	4.8	12.7	17.9	20.3	22.6	21.0	17.3	12.6	7.6	5.6
	2%	DB	1.6	2.2	6.0	17.8	24.2	27.4	28.8	26.1	20.3	12.6	6.7	4.0
		MCWB	1.0	1.1	2.7	10.3	16.2	19.3	20.7	19.2	15.9	10.5	5.9	3.4
	5%	DB	1.0	1.3	4.1	14.8	21.7	25.0	26.8	24.1	18.1	11.1	5.3	2.3
		MCWB	0.4	0.4	1.7	8.9	13.9	17.9	19.8	18.1	14.6	9.7	4.6	1.7
	10%	DB	0.1	0.5	2.7	11.6	19.4	22.7	24.8	22.0	16.2	9.7	4.1	1.2
		MCWB	-0.3	-0.2	1.1	6.9	12.7	16.7	18.8	17.1	13.4	8.6	3.6	0.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.0	2.4	5.4	12.8	19.3	21.8	23.0	22.2	18.1	13.0	8.0	5.6
		MCDB	2.3	3.0	8.5	20.7	24.7	27.2	29.4	28.0	21.8	15.8	8.3	6.0
	2%	WB	1.1	1.3	3.3	10.9	17.2	20.3	21.7	20.2	16.6	11.0	6.1	3.5
		MCDB	1.4	1.9	5.1	16.2	22.9	25.8	27.1	24.6	19.5	12.0	6.6	3.9
	5%	WB	0.5	0.6	2.2	9.2	15.3	18.8	20.6	18.9	15.1	9.8	4.7	1.8
		MCDB	0.9	1.2	3.4	13.7	19.4	23.6	25.4	22.8	17.4	10.8	5.2	2.3
	10%	WB	-0.3	-0.2	1.3	7.5	13.6	17.4	19.5	17.8	13.8	8.7	3.6	0.7
		MCDB	0.2	0.5	2.4	11.3	18.0	21.5	23.7	21.2	15.8	9.6	4.1	1.0
Mean Daily Temperature Range		MDBR	6.5	7.7	8.9	9.9	11.0	10.6	10.4	10.3	8.6	5.5	4.2	5.1
	5% DB	MCDBR	4.0	5.0	9.8	15.7	15.0	14.1	13.6	13.7	12.1	8.0	4.5	3.5
		MCWBR	4.1	4.6	7.1	9.0	7.3	6.7	6.3	7.3	7.6	5.9	4.2	3.5
		MCDBR	4.1	4.7	7.2	14.2	12.5	12.2	12.2	10.7	6.9	4.3	3.3	
	5% WB	MCWBR	4.3	4.5	5.5	8.5	6.7	6.3	6.0	6.8	7.2	5.4	4.2	3.4
Clear-Sky Solar Irradiance	taub	0.269	0.313	0.334	0.379	0.365	0.361	0.395	0.392	0.356	0.334	0.296	0.253	
	taud	2.062	2.060	2.108	2.208	2.408	2.439	2.363	2.351	2.431	2.371	2.167	2.028	
	Ebn at Noon	504	667	784	817	865	873	829	801	774	666	473	380	
	Edh at Noon	49	84	109	119	104	103	109	103	81	63	45	35	
All-Sky Solar Radiation	RadAvg	0.36	1.09	2.56	3.89	5.15	5.45	5.31	4.09	2.38	1.03	0.33	0.20	
	RadStd	0.06	0.23	0.35	0.46	0.34	0.34	0.66	0.28	0.26	0.18	0.06	0.03	